JEE Mathematics: Quick Review with Grading and Practice Map - Complex Number (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (98 questions; a question that appears in two of these is counted once): High = 9 or more questions, Medium = 1 to 8 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction to Complex Number

Question 01

What is the real part and imaginary part of a complex number and how are they represented?

Model Answer

$z = x + iy$ $(x, y \in \mathbb{R},\ i = \sqrt{-1})$: real part $\mathrm{Re}(z) = x$, imaginary part $\mathrm{Im}(z) = y$ (a real number, without i).

Ex: $z = 3 - 4i$: $\mathrm{Re}(z) = 3$, $\mathrm{Im}(z) = -4$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 18 | DPP: 1(11, 14) | Question Bank: 315, 341

Question 02

What is a purely real / purely imaginary / imaginary number?

Model Answer

Purely real: $\mathrm{Im}(z) = 0$; purely imaginary: $\mathrm{Re}(z) = 0,\ \mathrm{Im}(z) \neq 0$; imaginary (non-real): $\mathrm{Im}(z) \neq 0$. 0 is purely real and also taken as purely imaginary.

Ex: 5 is purely real, $-2i$ is purely imaginary, $1 + i$ is imaginary.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 1 | DPP: 1(1-2)

Integral Powers of Iota

Question 03

How to simplify a higher integral power of iota?

Model Answer

$i^2 = -1,\ i^3 = -i,\ i^4 = 1$; divide the power by 4: $i^{4k+r} = i^r$. Sum of four consecutive powers $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$.

Ex: $i^{103} = i^3 = -i;\ \sum_{n=1}^{13}(i^n + i^{n+1}) = i - 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 6; J-II 6; J-M 46 | DPP: 1(12) | Question Bank: 329

Also used in: Sheet: J-M 38 | DPP: 6(9) | Question Bank: 355

Argand Plane

Question 04

Can you plot any given complex number on the complex plane?

Model Answer

$z = x + iy$ is the point (x, y): real part on the x-axis (real axis), imaginary part on the y-axis (imaginary axis).

Ex: $-2 + 3i$ is the point $(-2, 3)$ in the second quadrant.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 05

Where is the position of a purely real / purely imaginary number on the complex plane?

Model Answer

Purely real numbers lie on the real (x) axis; purely imaginary numbers lie on the imaginary (y) axis.

Ex: $z = -7$ is $(-7, 0)$; $z = 4i$ is $(0, 4)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Algebra of Complex Number

Question 06

How to add / subtract / multiply and divide two or more complex numbers?

Model Answer

$$(a + ib) \pm (c + id) = (a \pm c) + i(b \pm d), \qquad (a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Two complex numbers are equal iff real parts and imaginary parts are equal separately.

Ex: $(2 + 3i)(1 - 2i) = 2 - 4i + 3i + 6 = 8 - i$.

Grade - High

Practice map, main concept: Sheet: J-I 7; J-II 13, 18; J-M 5, 22, 27, 31, 36, 49; C-II 5-7, 14, 17, 20 | DPP: 1(2, 4, 11, 15-16, 18); 2(4, 17) | Question Bank: 311, 334, 349, 353

Also used in: Sheet: J-II 9, 11; J-A 12; C-II 21 | DPP: 1(9, 17); 2(18); 5(14); 6(8) | Question Bank: 309

Question 07

What to do when a complex number is given in the denominator?

Model Answer

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{c^2 + d^2}$$

Ex: $\dfrac{1 + 2i}{2 - i} = \dfrac{(1 + 2i)(2 + i)}{5} = \dfrac{5i}{5} = i.$

Grade - Medium

Practice map, main concept: Sheet: J-M 7 | DPP: 1(9); 2(18) | Question Bank: 309, 315

Also used in: Sheet: J-I 6, 12; J-M 14, 24, 47; J-A 24 | DPP: 1(10, 15, 18); 3(14)

Question 08

In which type of numbers is inequality defined and in which is it not defined?

Model Answer

Inequalities ($<, >$) are defined only for real numbers; $z_1 > z_2$ has no meaning for non-real numbers. Moduli, real and imaginary parts are real, so $|z_1| > |z_2|$, $\text{Re}(z) \geq 0$ are allowed.

Ex: $3 + 2i > 1 + i$ is meaningless, but $|3 + 2i| > |1 + i|$ is true ($\sqrt{13} > \sqrt{2}$).

Grade - Low

Practice map, main concept: none

Also used in: DPP: 2(8); 3(11)

Question 09

How to solve an equation in z that also contains $\bar{z}$ or $|z|$ (e.g. $z^n = \bar{z}$, $z^2 + |z| = 0$)?

Model Answer

Either take modulus of both sides first (gives $|z|$), then use $z\bar{z} = |z|^2$ or arguments; or put $z = x+iy$ and equate real and imaginary parts. Do not forget $z = 0$.

Ex: $z^3 = \bar{z}$: $|z|^3 = |z| \Rightarrow |z| = 0$ or 1; for $|z| = 1$, $z^4 = z\bar{z} = 1$; so $z = 0, \pm 1, \pm i$: 5 solutions.

Grade - High

Practice map, main concept: Sheet: J-I 2, 16; J-II 5, 12; J-M 11-13, 21, 29, 40; J-A 18, 22, 25; C-I 14 | DPP: 1(1, 6); 3(16) | Question Bank: 323, 332, 337, 344

Also used in: Sheet: J-II 48; J-A 14 | DPP: 3(15) | Question Bank: 319

Square Root of a Complex Number

Question 10

How to find the square root of a complex number?

Model Answer

Let $\sqrt{a + ib} = x + iy$: $x^2 - y^2 = a$, $2xy = b$, $x^2 + y^2 = \sqrt{a^2 + b^2}$. Then x^2, y^2 follow; the sign of b fixes the signs of x, y (same signs if $b > 0$).

Ex: $\sqrt{9 + 40i} = \pm(5 + 4i)$; $\sqrt{-11 - 60i} = \pm(5 - 6i)$.

Grade - Low

Practice map, main concept: DPP: 2(16)

Also used in: Sheet: J-II 1, 10; C-I 10 | DPP: 1(3)

Question 11

How to solve a quadratic equation if its coefficients are complex numbers?

Model Answer

Use $z = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and find $\sqrt{b^2 - 4ac}$ as a complex square root. Roots need not be conjugates (that holds only for real coefficients).

Ex: $z^2 - (3 - 2i)z - (5i - 5) = 0$: $D = (3 - 2i)^2 + 4(5i - 5) = -15 + 8i = (1 + 4i)^2$, so $z = 2 + i$ or $1 - 3i$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; J-II 1, 10-11; C-I 10 | DPP: 1(3, 5); 3(4) | Question Bank: 320

Also used in: Sheet: J-A 1 | DPP: 3(6); 4(8)

Conjugate and its Properties

Question 12

How to find the conjugate of a complex number expressed as $a + ib$?

Model Answer

$\overline{a+ib} = a - ib$ (change the sign of the imaginary part). $z + \bar{z} = 2\,\mathrm{Re}(z)$, $z - \bar{z} = 2i\,\mathrm{Im}(z)$, $\bar{\bar{z}} = z$.

Ex: $\overline{-3+5i} = -3 - 5i$.

Grade - Low

Practice map, main concept: Sheet: J-II 8-9; C-II 21 | DPP: 1(14, 17)

Also used in: Sheet: C-II 14

Question 13

What does the conjugate of a complex number represent on the complex plane?

Model Answer

$\bar{z}$ is the mirror image of z in the real axis: $(x, y) \to (x, -y)$.

Ex: $z = 2 + 3i$ at $(2, 3)$, $\bar{z}$ at $(2, -3)$.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 2(1)

Question 14

How to prove / what to do when a given complex number is to be purely real?

Model Answer

z is purely real $\iff z = \bar{z} \iff \mathrm{Im}(z) = 0$. For a quotient, equate it to its conjugate and cross-multiply (or make Im $= 0$).

Ex: z non-real and $\dfrac{1 - z + z^2}{1 + z + z^2}$ real $\Rightarrow z\bar{z} = 1$, i.e. $|z| = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 24; J-II 2; J-M 16, 24; J-A 1, 21, 24; C-I 1, 19 | DPP: 1(13); 3(5, 10) | Question Bank: 308, 339

Also used in: Sheet: J-II 36; J-M 22; C-II 32 | Question Bank: 312

Question 15

How to prove / what to do when a given complex number is to be purely imaginary?

Model Answer

z is purely imaginary $\iff z + \bar{z} = 0 \iff \mathrm{Re}(z) = 0$ (with $z \neq 0$).

Ex: $\dfrac{z-1}{z+1}$ purely imaginary $\Rightarrow \dfrac{z-1}{z+1} + \dfrac{\bar{z}-1}{\bar{z}+1} = 0 \Rightarrow z\bar{z} = 1$, $|z| = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 12, 17; J-M 14-15, 47; C-I 8 | Question Bank: 312, 338

Also used in: Sheet: J-I 26; J-II 37; J-M 48 | DPP: 1(13); 3(5); 5(5) | Question Bank: 330

Question 16
How to take the conjugate of a complex number expressed as a sum / product of several numbers?

Model Answer

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \, \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad \overline{z^n} = (\bar{z})^n$$

Ex: $z^5 + (\bar{z})^5 = z^5 + \overline{z^5} = 2\,\mathrm{Re}(z^5)$.

Grade - Low
Practice map, main concept: Sheet: C-I 3
Also used in: none

Modulus and its Properties

Question 17
How to find the modulus of a complex number expressed as $a + ib$?

Model Answer

$$|a + ib| = \sqrt{a^2 + b^2} \ (\geq 0)$$

Ex: $|3 - 4i| = 5$.

Grade - Medium
Practice map, main concept: DPP: 2(8)
Also used in: Sheet: J-M 10, 13, 27; J-A 23; C-I 8 | DPP: 1(6)

Question 18
What does the modulus of a complex number represent on the complex plane?

Model Answer
$|z|$ = distance of the point z from the origin.
Ex: all z with $|z| = 2$ form the circle of radius 2 about O.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 19
What is the relation between z, its conjugate and $|z|$?

Model Answer

$$z\,\bar{z} = |z|^2, \qquad |z| = |\bar{z}| = |-z|, \qquad |z| = 1 \iff \bar{z} = \frac{1}{z}$$

Ex: $z = 1 + 2i$: $z\bar{z} = 1 + 4 = 5 = |z|^2$.

Grade - Medium
Practice map, main concept: DPP: 2(1) | Question Bank: 316
Also used in: Sheet: J-I 31; J-II 5; J-A 25

Question 20
How to take the modulus of a complex number expressed as a product of several numbers?

Model Answer

$$|z_1 z_2 \cdots z_n| = |z_1||z_2| \cdots |z_n|, \qquad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \qquad |z^n| = |z|^n$$

Ex: $\left|\dfrac{(1+i)^6}{(1-i\sqrt{3})^3}\right| = \dfrac{(\sqrt{2})^6}{2^3} = 1.$

Grade - Medium

Practice map, main concept: Sheet: J-I 3 | DPP: 3(2, 8); 6(8)
Also used in: Sheet: J-II 12; J-M 31; C-I 9; C-II 7 | DPP: 3(17) | Question Bank: 334

Question 21
What is the difference between the square of z and the square of $|z|$?

Model Answer
$z^2 = a^2 - b^2 + 2iab$ is complex; $|z|^2 = a^2 + b^2 = z\bar{z}$ is real. $z^2 = |z|^2$ only when z is real.
Ex: $z = 1 + i$: $z^2 = 2i$ but $|z|^2 = 2$.

Grade - Low

Practice map, main concept: none
Also used in: none

Question 22
What is the difference in finding the conjugate and the modulus of a complex number expressed as a sum of several numbers?

Model Answer
Conjugate distributes over a sum: $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$; modulus does not: $|z_1 + z_2| \leq |z_1| + |z_2|$ (find the sum first, then its modulus).
Ex: $|1 + i| = \sqrt{2}$, but $|1| + |i| = 2$.

Grade - Low

Practice map, main concept: none
Also used in: none

Question 23
If two complex numbers are equal, then what is the relation between their moduli?

Model Answer
$z_1 = z_2 \Rightarrow |z_1| = |z_2|$ (the converse is false). Taking modulus of both sides of an equation is often the first step.
Ex: $|3 + 4i| = |5|$ but $3 + 4i \neq 5$.

Grade - Low

Practice map, main concept: none
Also used in: Sheet: J-I 2; C-I 14 | DPP: 1(5); 3(16)

Argument and its Properties

Question 24

How to find the argument of a complex number expressed as $a + ib$?

Model Answer

Find $\alpha = \tan^{-1}\left|\frac{b}{a}\right|$ and use the quadrant: 1st: α, 2nd: $\pi - \alpha$, 3rd: $-(\pi - \alpha)$, 4th: $-\alpha$ (principal value).

Ex: $\arg(-1 + i\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$; $\arg(-1 - i) = -\frac{3\pi}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 1 | DPP: 1(10)

Also used in: Sheet: J-I 29; J-M 7, 17; J-A 15 | Question Bank: 340

Question 25

What is the difference between argument and amplitude (principal value of argument)?

Model Answer

Argument = any angle θ of $\vec{Oz}$ with the positive real axis (many values, differing by 2π); amplitude / principal argument = the one value in $(-\pi, \pi]$.

Ex: $z = i$: arguments $\frac{\pi}{2} + 2k\pi$, principal argument $\frac{\pi}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-A 13 | DPP: 3(23)

Also used in: Sheet: J-M 28 | DPP: 2(5, 19-20)

Question 26

How to find the general value of the argument?

Model Answer

$$\mathrm{Arg}(z) = \arg(z) + 2k\pi, \quad k \in \mathbb{Z}$$

Ex: $z = 1 - i$: general argument $-\frac{\pi}{4} + 2k\pi$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 35

Question 27

What does the argument of a complex number represent on the complex plane?

Model Answer

The angle made by the line from O to z with the positive real axis (anticlockwise positive). $\arg(z) = \theta$ (fixed) is a ray from O (origin excluded).

Ex: $\arg z = -\frac{\pi}{3}$: ray in the fourth quadrant making 60° below the x-axis.

Grade - Low

Practice map, main concept: Sheet: J-II 20, 41, 45; C-I 18 | DPP: 2(5, 15)

Also used in: Sheet: C-I 13 | DPP: 2(14); 3(13)

Question 28
If two complex numbers are equal, then what is the relation between their arguments?

Model Answer
Equal principal arguments; general arguments differ by $2k\pi$. A complex number is fixed by modulus and argument together.
Ex: $2e^{i\pi/3} = re^{i\theta} \Rightarrow r = 2,\ \theta = \dfrac{\pi}{3} + 2k\pi.$

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 37

Question 29
What is the argument of complex numbers lying on the real and imaginary axes?

Model Answer
Positive real: 0; negative real: π; positive imaginary: $\dfrac{\pi}{2}$; negative imaginary: $-\dfrac{\pi}{2}$; $\arg(0)$ is not defined.
Ex: $\arg(-5) = \pi$, $\arg(-3i) = -\dfrac{\pi}{2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Polar Form and Exponential Form

Question 30
What is the polar form? Explain all its components.

Model Answer

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z|,\ \theta = \arg(z)$$

$r \geq 0$ is the modulus and θ the argument (any value; the principal one is usual).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 31
How to express a complex number in polar form if it is given in $a + ib$ form?

Model Answer
Find $r = \sqrt{a^2 + b^2}$ and $\theta = \arg(a + ib)$ (by quadrant), then write $r(\cos\theta + i\sin\theta)$.
Ex: $-1 + i = \sqrt{2}\left(\cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4}\right).$

Grade - Medium
Practice map, main concept: DPP: 4(1)
Also used in: Sheet: J-M 1, 8, 37; C-I 16 | DPP: 2(12); 3(19) | Question Bank: 316, 354

Question 32
What does $\mathrm{CiS}(\theta)$ mean?

Model Answer

$$\mathrm{CiS}\,\theta = \cos\theta + i\sin\theta = e^{i\theta}, \qquad \mathrm{CiS}\,\alpha \cdot \mathrm{CiS}\,\beta = \mathrm{CiS}(\alpha+\beta)$$

Ex: $2\,\mathrm{CiS}\dfrac{\pi}{2} = 2i$.

Grade - Medium

Practice map, main concept: Sheet: J-A 27 | DPP: 2(7)

Also used in: none

Question 33
What is the exponential form? Explain all its components.

Model Answer
$z = re^{i\theta}$ with Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$; $r = |z|$, $\theta = \arg z$.
$\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2}$, $\sin\theta = \dfrac{e^{i\theta}-e^{-i\theta}}{2i}$; $|e^{i\theta}| = 1$.

Grade - Medium

Practice map, main concept: Sheet: C-I 21 | DPP: 3(15) | Question Bank: 319

Also used in: Sheet: J-A 20 | DPP: 2(7); 5(1)

Question 34
How to express a complex number in exponential form if it is given in $a + ib$ form?

Model Answer
Find r and θ exactly as for the polar form, then write $re^{i\theta}$.
Ex: $1 - i\sqrt{3} = 2e^{-i\pi/3}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 35
When to use the polar form in questions?

Model Answer
When the question needs $\cos, \sin$ of the argument explicitly: real / imaginary parts of products and powers, comparing with $a + ib$, trigonometric conditions.
Ex: $(1+i)^4 = (\sqrt{2})^4(\cos\pi + i\sin\pi) = -4$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 36
When to use the exponential form in questions?

Model Answer
For products, quotients, powers and roots (arguments add / subtract / multiply), for $|z| = 1$ ($z = e^{i\theta}$) and for rotation.
Ex: $\dfrac{(1+i)^8}{(1-i)^6} = \dfrac{16e^{2\pi i}}{8e^{-3\pi i/2}} = 2e^{i7\pi/2} = -2i$.

Grade - Medium

Practice map, main concept: Sheet: J-I 26; J-II 31; J-A 23; C-II 2

Also used in: none

Question 37

When will we prefer the polar form, and when the exponential form, over the other form?

Model Answer

Exponential: algebra of many factors, powers, roots, $z = e^{i\theta}$ substitutions, rotation. Polar: final answer in cos / sin, or when real and imaginary parts must be compared.

Ex: $|z| = 1 \Rightarrow \dfrac{z}{1+z^2} = \dfrac{1}{e^{-i\theta}+e^{i\theta}} = \dfrac{1}{2\cos\theta}$ (real).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 38

How to find the argument of a complex number expressed as a product / quotient of several numbers?

Model Answer

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2 + 2k\pi, \qquad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 + 2k\pi$$

Choose k so that the result lies in $(-\pi, \pi]$. Ex: $\arg\left((1+i\sqrt{3})(1+i)\right) = \dfrac{\pi}{3} + \dfrac{\pi}{4} = \dfrac{7\pi}{12}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 8, 38 | DPP: 2(13); 3(17, 19); 4(3); 5(4); 6(2) | Question Bank: 355

Also used in: DPP: 3(8); 4(9)

Question 39

How to find the argument of the conjugate of a complex number?

Model Answer

$\arg(\bar{z}) = -\arg(z)$ (z not negative real); $\arg(-z) = \arg z \pm \pi$.

Ex: $\arg(\overline{1+i}) = -\dfrac{\pi}{4}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 4-5 | DPP: 1(7)

Question 40

What is the pattern of the properties of argument?

Model Answer

Argument behaves like a logarithm: product $\to$ sum, quotient $\to$ difference, power $\to$ multiple, conjugate $\to$ negative (each up to $2k\pi$).

Ex: $|z| = |w|$, $\arg z + \arg w = \pi \Rightarrow zw = -|z|^2 = -z\bar{z} \Rightarrow z = -\bar{w}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 4-5, 35 | DPP: 1(7)

Also used in: Sheet: J-I 10; J-A 13 | DPP: 4(3); 6(2)

Question 41

How to convert look-alike forms of the polar form into the correct polar form?

Model Answer

$\cos\theta - i\sin\theta = e^{-i\theta}$; $\sin\theta + i\cos\theta = e^{i(\pi/2-\theta)}$; $-\cos\theta - i\sin\theta = e^{i(\pi+\theta)}$; $1+\cos\theta+i\sin\theta = 2\cos\dfrac{\theta}{2}\,e^{i\theta/2}$; $1 + i\tan\theta = \sec\theta\, e^{i\theta}$.

Check the sign of the modulus: $1+\cos\dfrac{10\pi}{9}+i\sin\dfrac{10\pi}{9} = 2\cos\dfrac{5\pi}{9}e^{i5\pi/9} = -2\cos\dfrac{5\pi}{9}\,e^{-i4\pi/9}$, so $|z| = -2\cos\dfrac{5\pi}{9}$, $\arg z = -\dfrac{4\pi}{9}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 17, 42; J-M 2, 28; C-I 6 | DPP: 2(19-20)

Also used in: Sheet: J-I 1 | DPP: 3(23)

Question 42

When $|z| = 1$, what to do when $(1+z)$ or $(1-z)$ is used in questions?

Model Answer

$$z = e^{i\theta}: \quad 1+z = 2\cos\frac{\theta}{2}\,e^{i\theta/2}, \qquad 1-z = -2i\sin\frac{\theta}{2}\,e^{i\theta/2}$$

Hence $\dfrac{1+z}{1+\bar{z}} = z$, $\arg\dfrac{1+z}{1+\bar{z}} = \theta$, and $\dfrac{1-z}{1+z} = -i\tan\dfrac{\theta}{2}$ (purely imaginary).

Grade - Low

Practice map, main concept: Sheet: J-I 10, 31; C-II 31-32

Also used in: DPP: 3(5)

Geometrical Interpretation

Question 43

What is the distance formula for the complex plane?

Model Answer

$$|z_1 - z_2| = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

Ex: distance between $1+2i$ and $4-2i$ is $|-3+4i| = 5$.

Grade - Medium

Practice map, main concept: Sheet: C-II 24

Also used in: Sheet: J-II 14, 16; J-A 18 | DPP: 3(4); 6(7)

Question 44

What does $z_2 - z_1$ represent on the complex plane (similar to a vector)?

Model Answer

$z_2 - z_1$ is the vector $\vec{PQ}$ from $P(z_1)$ to $Q(z_2)$: length $|z_2 - z_1|$, direction $\arg(z_2 - z_1)$.

Ex: $P(1+i)$, $Q(3+3i)$: $\vec{PQ} = 2+2i$, length $2\sqrt{2}$, direction $\dfrac{\pi}{4}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 45

What does $|z_1 - z_2|$ represent on the complex plane?

Model Answer

The distance between the points z_1 and z_2. So $|z - z_0|$ with z on a circle (centre c, radius r) has max $|z_0 - c| + r$ and min $\big||z_0 - c| - r\big|$.

Ex: $|z + 4| \le 3$: $|z + 1|$ has max $3 + 3 = 6$, min 0 (as -1 lies inside the disc).

Grade - High

Practice map, main concept: Sheet: J-I 8, 23; J-II 30; J-M 19; J-A 5, 15, 20; C-I 12 | DPP: 2(10); 4(4) | Question Bank: 322, 342

Also used in: Sheet: J-I 25; J-II 29; J-M 32-33, 39, 48; C-II 25 | DPP: 3(7) | Question Bank: 308, 330, 335-336, 356

Question 46

What is the expression of the locus which represents a circle?

Model Answer

$|z - z_0| = r$: circle; $|z - z_0| < r$: inside; $r_1 < |z - z_0| < r_2$: annulus (ring).

Ex: $|2z - 1| = 3 \iff \left|z - \frac{1}{2}\right| = \frac{3}{2}$: centre $\left(\frac{1}{2}, 0\right)$, radius $\frac{3}{2}$.

Grade - High

Practice map, main concept: DPP: 2(3); 3(11); 4(15)

Also used in: Sheet: J-I 8, 11, 23; J-II 20, 25-27, 30, 45; J-M 35, 41, 43-44, 50; J-A 6, 11, 17, 28; C-I 18, 22; C-II 15 | DPP: 2(10, 15); 3(22); 4(1, 4) | Question Bank: 317, 322, 324, 326-327, 345, 350, 352

Question 47

Relate all the locus results of straight line based on the distance formula to their application in complex numbers.

Model Answer

$|z-z_1| = |z-z_2|$: perpendicular bisector; $\left|\frac{z - z_1}{z - z_2}\right| = k \neq 1$: circle (Apollonius); $|z-z_1|+|z-z_2| = k$: ellipse if $k > |z_1 - z_2|$, segment if $k = |z_1 - z_2|$, empty if smaller.

$\big||z - z_1| - |z - z_2|\big| = k$: hyperbola if $k < |z_1 - z_2|$, two rays if equal. Ex: $|z - 4| + |z + 4| = 16$: ellipse $\frac{x^2}{64} + \frac{y^2}{48} = 1$.

Grade - High

Practice map, main concept: Sheet: J-I 21-22; J-II 22-23, 27, 29, 38, 51; J-M 4, 9-10, 26, 50; J-A 28; C-I 4, 24 | DPP: 2(2, 11); 3(21-22); 5(2, 7) | Question Bank: 318, 331, 350

Also used in: Sheet: J-II 39-40, 49; J-M 6; C-II 8 | DPP: 6(6) | Question Bank: 348

Question 48

What is the section formula for internal and external division in the complex plane, and its similarity with vectors and straight line?

Model Answer

Internal: $z = \frac{mz_2 + nz_1}{m + n}$; external: $z = \frac{mz_2 - nz_1}{m - n}$.

Same as for position vectors / coordinates. Ex: midpoint of $2 + 4i$ and $6 - 2i$ is $4 + i$.

Grade - Low

Practice map, main concept: Sheet: J-II 44; C-II 1

Also used in: none

Question 49

What are the results for the centroid / incentre / excentres / orthocentre / circumcentre of a triangle?

Model Answer

$$G = \frac{z_1 + z_2 + z_3}{3}, \quad I = \frac{az_1 + bz_2 + cz_3}{a + b + c}, \quad I_1 = \frac{-az_1 + bz_2 + cz_3}{-a + b + c}$$

$$H = \frac{\sum z_1 \tan A}{\sum \tan A}, \qquad O = \frac{\sum z_1 \sin 2A}{\sum \sin 2A}$$

If the circumcentre is at the origin, orthocentre $= z_1 + z_2 + z_3$. Also $\sum |z - z_k|^2$ is least at $z = G$.

Grade - Low

Practice map, main concept: Sheet: J-II 14; C-I 2 | DPP: 3(14)

Also used in: Sheet: J-II 24

Question 50

What do 0, z_1, z_2 and $z_1 + z_2$ represent on a complex plane, and what is the importance of $z_1 - z_2$ in this situation?

Model Answer

Vertices of a parallelogram; $z_1 + z_2$ and $z_1 - z_2$ are its diagonals. $|z_1 + z_2| = |z_1 - z_2| \Rightarrow$ rectangle ($\arg \frac{z_1}{z_2} = \pm\frac{\pi}{2}$); $|z_1| = |z_2| \Rightarrow$ rhombus.

Points z_1, z_2, z_3, z_4 (in order) form a parallelogram iff $z_1 + z_3 = z_2 + z_4$ (diagonals bisect).

Grade - Low

Practice map, main concept: Sheet: J-I 15 | DPP: 2(6); 3(18); 5(9)

Also used in: none

Question 51

How to find the equation of the locus of a point satisfying given conditions in a complex plane, by putting $z = x + iy$?

Model Answer

Put $z = x + iy$ ($\bar{z} = x - iy$), simplify the condition into a real equation in x, y, and identify the curve (line, circle, conic).

Ex: $|z| = \mathrm{Re}(z) + 2$: $\sqrt{x^2 + y^2} = x + 2 \Rightarrow y^2 = 4x + 4$ (parabola).

Grade - High

Practice map, main concept: Sheet: J-I 18, 20; J-II 26, 36-37, 39-40; J-M 3, 6, 45, 48; J-A 4, 11-12; C-I 17, 22; C-II 23 | DPP: 3(20) | Question Bank: 313, 328, 330

Also used in: Sheet: J-I 14, 16, 24; J-II 41, 51; J-M 11, 15-16, 25; J-A 5; C-II 24 | DPP: 3(10) | Question Bank: 338-339

Question 52

How to solve questions on the locus of a point satisfying given conditions in a complex plane, by relating with coordinate geometry?

Model Answer

Read each modulus as a distance and each argument as an angle; recognise the standard figure (circle, bisector, conic, arc) and use its geometry (tangent, perpendicular distance, extreme points) without expanding.

Ex: $|z_1| = 12$, $|z_2 - 3 - 4i| = 5$: second circle (centre at distance 5, radius 5) lies inside the first, so $\min |z_1 - z_2| = 12 - 10 = 2$.

Grade - Low

Practice map, main concept: Sheet: C-II 10

Also used in: none

Question 53

Explain the triangle inequality for complex numbers.

Model Answer

$$\big||z_1| - |z_2|\big| \le |z_1 \pm z_2| \le |z_1| + |z_2|$$

Ex: $|z| \ge 2 \Rightarrow \left|z + \frac{1}{2}\right| \ge |z| - \frac{1}{2} \ge \frac{3}{2}$, equality at $z = -2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 25; J-II 53; J-M 33, 42; J-A 17; C-I 5; C-II 4, 25 | DPP: 4(2) | Question Bank: 325, 336

Also used in: Sheet: C-I 4, 12 | DPP: 3(1)

Question 54

What is the situation if we have equality in the triangle inequality?

Model Answer

$|z_1 + z_2| = |z_1| + |z_2| \iff O, z_1, z_2$ collinear with z_1, z_2 on the same side of O ($\arg z_1 = \arg z_2$);
$|z_1 + z_2| = \big||z_1| - |z_2|\big| \iff$ opposite sides ($\arg z_1 - \arg z_2 = \pi$).
Also $|z - z_1| + |z - z_2| = |z_1 - z_2| \iff z$ lies on the segment $z_1 z_2$.

Grade - Low

Practice map, main concept: DPP: 3(1, 7); 5(5)

Also used in: Sheet: J-II 43, 47, 53

Question 55

Use of $z\bar{z} = |z|^2$ in questions.

Model Answer

Square a modulus as a product: $|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\,\mathrm{Re}(z_1\bar{z}_2)$; this proves identities and turns $|z| = r$ into $\bar{z} = \frac{r^2}{z}$.

Ex: $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$; $|1 - z_1\bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 11, 14; J-II 3-4, 19; J-M 35, 44; J-A 6; C-I 9, 20 | DPP: 3(3); 6(7) | Question Bank: 327, 352

Also used in: Sheet: J-I 17; J-II 34; J-M 23, 42; J-A 16, 21; C-I 19; C-II 18, 22 | DPP: 6(15) | Question Bank: 325

Angles

Question 56

Relate $\arg(z)$ with a vector.

Model Answer

$\arg z$ is the direction of the vector $\vec{Oz}$; unit vector along it is $\frac{z}{|z|}$. Sum of unit vectors gives the angle bisector.

Ex: bisector of the angle between $3 + 4i$ and $5i$ has direction $\frac{3+4i}{5} + i = \frac{3+9i}{5}$.

Grade - Low

Practice map, main concept: DPP: 4(7)

Also used in: none

Question 57

How to write any angle of a line on the complex plane as $\arg(z_2 - z_1)$?

Model Answer

The line from $A(z_1)$ to $B(z_2)$ makes angle $\arg(z_2 - z_1)$ with the positive real axis. $\arg(z - z_1) = \theta$ is a ray from z_1 (excluding z_1).

Ex: $\arg(z - 2 - 3i) = \dfrac{\pi}{4}$: ray $y = x + 1$, $x > 2$.

Grade - Low

Practice map, main concept: Sheet: C-I 13 | DPP: 2(14)

Also used in: none

Question 58

How to write any angle between two lines on the complex plane as $\arg\left(\dfrac{z_3 - z_1}{z_2 - z_1}\right)$?

Model Answer

Angle from $\vec{AB}$ to $\vec{AC}$ (anticlockwise) $= \arg(z_3 - z_1) - \arg(z_2 - z_1) = \arg\dfrac{z_3 - z_1}{z_2 - z_1}$.

Ex: $A = 0$, $B = 1$, $C = 1 + i$: $\angle BAC = \arg(1 + i) = \dfrac{\pi}{4}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 2 | DPP: 5(4)

Question 59

What is the use of the initial and final vector in writing an angle?

Model Answer

$\arg\left(\dfrac{\text{final vector}}{\text{initial vector}}\right)$ = angle turned anticlockwise from the initial to the final vector.

Both vectors must start from the vertex of the angle. Ex: angle at z_1 from $z_1\vec{z}_2$ to $z_1\vec{z}_3$ is $\arg\dfrac{z_3 - z_1}{z_2 - z_1}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 60

If the angle is measured in the clockwise direction, then what to do?

Model Answer

Take it negative: a clockwise angle θ is $-\theta$, i.e. $\arg\dfrac{\text{final}}{\text{initial}} = -\theta$ (or interchange initial and final).

Ex: rotating $z_1 = 3 + 5i$ clockwise by $\dfrac{\pi}{2}$ gives $z_1 e^{-i\pi/2} = 5 - 3i$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 61
What to keep in mind while marking the angle between two line segments?

Model Answer
Both segments must point away from the common vertex, and the sense (anticlockwise +, clockwise −) must be fixed; otherwise the angle becomes $\pi - \theta$ or $-\theta$.

Ex: $\arg\dfrac{z_1 - z}{z_2 - z}$ is the angle at z, not at z_1.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 62
What is the condition for two lines to be parallel?

Model Answer
$AB \parallel CD \iff \dfrac{z_4 - z_3}{z_2 - z_1}$ is purely real.

Collinear points: $\dfrac{z_3 - z_1}{z_2 - z_1}$ real. Ex: $1 + i,\ 2 + 2i,\ 4 + 4i$ are collinear (ratio 3).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 63
What is the condition for two lines to be perpendicular?

Model Answer
$AB \perp CD \iff \dfrac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary.

Ex: $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Rightarrow \mathrm{Re}(z_1\bar{z}_2) = 0 \Rightarrow \dfrac{z_1}{z_2}$ purely imaginary ($Oz_1 \perp Oz_2$).

Grade - Medium

Practice map, main concept: Sheet: J-II 24

Also used in: Sheet: J-II 50; J-M 3; C-I 20 | DPP: 2(6); 3(18); 5(9)

Question 64
Application of the locus of P when the angle between PA and PB is fixed, in complex numbers.

Model Answer
$\arg\dfrac{z - z_1}{z - z_2} = \theta$ $(0 < \theta < \pi)$: arc of a circle through z_1, z_2 (endpoints excluded); $\theta = \pm\dfrac{\pi}{2}$: semicircle on diameter z_1z_2; $\theta = 0$ or π: parts of the line z_1z_2.

Ex: $\arg\dfrac{z - 1}{z + 1} = \dfrac{\pi}{4}$: chord of length 2 subtends 90° at the centre: centre $(0, 1)$, radius $\sqrt{2}$ (upper arc).

Grade - Medium

Practice map, main concept: Sheet: J-A 19 | DPP: 5(8)

Also used in: Sheet: J-II 23, 52

Rotation

Question 65

How to use the result of rotation when two line segments are given?

Model Answer

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\theta}$$

Here $\theta = \angle BAC$ measured anticlockwise from AB to AC.

Ex: square $ABCD$ (anticlockwise), $A = z_1$, $B = z_2$: $D = z_1 + i(z_2 - z_1)$, $C = z_2 + i(z_2 - z_1)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 27, 32, 34; C-II 12, 15 | DPP: 3(9); 4(6, 8-9) | Question Bank: 321

Also used in: Sheet: J-II 21, 28; C-II 10

Question 66

What happens to a vector in the complex plane if we multiply it by $e^{i\theta}$?

Model Answer

It rotates through θ anticlockwise about the origin, length unchanged ($|e^{i\theta}| = 1$). Multiplying by $ke^{i\theta}$ also stretches it k times.

Ex: $-4 + 5i$ turned through 180° and stretched 1.5 times: $1.5(-1)(-4 + 5i) = 6 - \frac{15}{2}i$.

Grade - Medium

Practice map, main concept: Sheet: J-I 30, 33

Also used in: Question Bank: 321

Question 67

What happens to a vector in the complex plane if we multiply it by i?

Model Answer

$i = e^{i\pi/2}$: rotation through 90° anticlockwise; $-i$: 90° clockwise.

Ex: $i(5 + 4i) = -4 + 5i$.

Grade - Medium

Practice map, main concept: Sheet: J-M 17 | Question Bank: 340

Also used in: Sheet: J-I 27; C-I 3; C-II 12 | DPP: 3(9)

Question 68

What is the relation between z_1, z_2 and z_3 if these are vertices of an equilateral triangle?

Model Answer

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1 \iff \frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$$

Equivalently $z_1 + \omega z_2 + \omega^2 z_3 = 0$ (or with ω, ω^2 swapped); centroid = circumcentre z_0 gives $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$. Roots of $z^3 - 3\alpha z^2 + 3\beta z + \gamma = 0$ form one iff $\alpha^2 = \beta$.

Grade - Medium

Practice map, main concept: Sheet: J-II 21, 46; C-I 25 | DPP: 4(5); 5(10) | Question Bank: 310

Also used in: Sheet: J-I 28, 32

Question 69
How to find the remaining vertices of a regular polygon of n sides if one vertex and its circumcentre are known?

Model Answer

$$z_k = z_0 + (z_1 - z_0)\, e^{i2\pi(k-1)/n}, \qquad k = 1, 2, \ldots, n$$

Ex: equilateral triangle in $|z| = 2$ with $z_1 = 1 + i\sqrt{3}$: $z_2 = z_1 e^{2\pi i/3} = -2$, $z_3 = 1 - i\sqrt{3}$.

Grade - Low
Practice map, main concept: Sheet: J-I 28-29; J-II 28
Also used in: Sheet: J-I 34

De-Moivre's Theorem (DMT)

Question 70
How to find an integral power z^n (or $\alpha^n + \beta^n$ for complex roots) using DMT?

Model Answer

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta, \qquad (re^{i\theta})^n = r^n e^{in\theta}\ (n \in \mathbb{Z})$$

Write each number (or each root of the quadratic) in exponential form, raise the power, then reduce $n\theta$ by multiples of 2π.

Ex: roots of $x^2 - \sqrt{2}x + 2 = 0$ are $\sqrt{2}e^{\pm i\pi/3}$, so $\alpha^{14} + \beta^{14} = 2 \cdot 2^7 \cos\dfrac{14\pi}{3} = -128$; and $(\sqrt{3}+i)^{100} = 2^{100}e^{i50\pi/3} = 2^{99}(-1 + i\sqrt{3})$.

Grade - Medium
Practice map, main concept: Sheet: J-I 37; J-M 1, 20, 37; C-I 15-16 | DPP: 2(12); 5(1); 6(9) | Question Bank: 343, 354
Also used in: Sheet: J-I 13; J-II 13, 17, 31, 42; J-M 2; C-I 6, 21 | Question Bank: 320

Question 71
What are the steps to find the nth root of a complex number using DMT?

Model Answer
(i) Write $z = re^{i\theta}$; (ii) replace θ by $\theta + 2k\pi$; (iii) $z^{1/n} = r^{1/n}e^{i(\theta+2k\pi)/n}$; (iv) put $k = 0, 1, \ldots, n-1$ (n distinct roots on the circle of radius $r^{1/n}$).
Ex: $\omega^3 = 8i = 8e^{i\pi/2}$: $\omega = 2e^{i(\pi/6+2k\pi/3)}$; second-quadrant root ($k = 1$): $2e^{i5\pi/6} = -\sqrt{3} + i$.

Grade - Low
Practice map, main concept: Sheet: J-I 36, 38; J-II 48; C-II 8 | DPP: 5(3, 6); 6(1, 6, 10)
Also used in: Sheet: C-I 24; C-II 5

Question 72
Why do we use the general value of the argument in place of the argument while using DMT for the nth root?

Model Answer
$e^{i\theta} = e^{i(\theta+2k\pi)}$, but after dividing by n the angles $\dfrac{\theta + 2k\pi}{n}$ give different points; with only θ we would get just one of the n roots.
Ex: $1^{1/3}$ with $\theta = 0$ gives only 1; with $2k\pi$ we get $1, \omega, \omega^2$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 38

Question 73
How to solve questions where it is given that an equation has a purely real / purely imaginary solution?

Model Answer
Put $z = x$ (real) or $z = iy$; separate real and imaginary parts; both must vanish, so the real x (or y) is a common root of two real equations.
Ex: $z^2 - (3+i)z + m + 2i = 0$ has a real root x: $x^2 - 3x + m = 0$ and $-x + 2 = 0 \Rightarrow x = 2,\ m = 2$.

Grade - Low
Practice map, main concept: Sheet: J-II 7, 15-16; C-I 11 | DPP: 2(9); 3(6)
Also used in: none

Cube Roots of Unity

Question 74
What are the three cube roots of unity and how are they represented?

Model Answer

$$z^3 = 1:\ 1,\ \omega = \frac{-1+i\sqrt{3}}{2} = e^{i2\pi/3},\ \omega^2 = \frac{-1-i\sqrt{3}}{2} = e^{-i2\pi/3} = \bar{\omega}$$

Ex: $\omega \cdot \omega^2 = \omega^3 = 1$ and $|\omega| = 1$.

Grade - Low
Practice map, main concept: Sheet: J-I 40
Also used in: Sheet: J-I 19, 42

Question 75
What is the sum of all three cube roots of unity?

Model Answer

$$1 + \omega + \omega^2 = 0$$

So $1 + \omega = -\omega^2$, $1 + \omega^2 = -\omega$. Ex: $(1 + \omega - \omega^2)(1 + \omega^2 - \omega) = (-2\omega^2)(-2\omega) = 4$.

Grade - Medium
Practice map, main concept: Sheet: J-I 39
Also used in: Sheet: J-I 41; J-M 34; C-II 29-30 | DPP: 4(13); 6(12) | Question Bank: 351

Question 76
How to find higher integral powers of ω?

Model Answer
Divide the power by 3: $\omega^{3k} = 1$, $\omega^{3k+1} = \omega$, $\omega^{3k+2} = \omega^2$.
Ex: $(1+\omega)^7 = (-\omega^2)^7 = -\omega^{14} = -\omega^2 = 1 + \omega$, so $A = B = 1$.

Grade - High
Practice map, main concept: Sheet: J-I 41-42; J-II 33, 52; J-M 30, 34; J-A 8, 10, 29; C-II 3, 29-30 | DPP: 4(10, 13-14); 5(14); 6(12, 14) | Question Bank: 314, 333, 348, 351
Also used in: Sheet: J-I 39, 43; J-A 3 | DPP: 4(11-12); 6(13) | Question Bank: 310, 346

Question 77
How to find the sum of any r^{th} power of all cube roots of unity?

Model Answer
$1+\omega^r+\omega^{2r}=3$ if r is a multiple of 3, otherwise 0.
Ex: $\omega^n+\omega^{2n}=-1$ for $n=3k+2$.

Grade - Medium
Practice map, main concept: Sheet: J-A 3 | DPP: 4(12); 6(13) | Question Bank: 346
Also used in: Sheet: J-A 10

Question 78
What do all three cube roots of unity represent on the complex plane, and can you find the dimensions of that triangle?

Model Answer
Vertices of an equilateral triangle inscribed in $|z|=1$ (centroid at O): side $\sqrt{3}$, area $\dfrac{3\sqrt{3}}{4}$.
Ex: $|1-\omega|=\left|\dfrac{3}{2}-\dfrac{\sqrt{3}}{2}i\right|=\sqrt{3}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 46

Question 79
What happens to a vector in the complex plane if we multiply it by ω?

Model Answer
$\omega=e^{i2\pi/3}$: rotation through 120° anticlockwise (by ω^2: 120° clockwise), length unchanged.
Ex: $\dfrac{b+c\omega+a\omega^2}{a+b\omega+c\omega^2}=\omega^2$ since $\omega^2(a+b\omega+c\omega^2)=a\omega^2+b+c\omega$.

Grade - Low
Practice map, main concept: DPP: 4(11)
Also used in: DPP: 4(6)

Question 80
Write all factors which are to be remembered.

Model Answer

$$x^2+x+1=(x-\omega)(x-\omega^2),\qquad x^2-x+1=(x+\omega)(x+\omega^2)$$
$$a^3+b^3+c^3-3abc=(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$$
$$a^2+b^2+c^2-ab-bc-ca=(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)=|a+b\omega+c\omega^2|^2\ (a,b,c\in\mathbb{R})$$

Ex: distinct integers a,b,c: $\min|a+b\omega+c\omega^2|^2=\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]_{\min}=\frac{1}{2}(1+1+4)=3$.

Grade - Medium
Practice map, main concept: Sheet: J-II 32; J-A 16; C-II 22 | DPP: 6(15)
Also used in: Sheet: J-II 33; J-M 30; C-II 11 | Question Bank: 333

Question 81
Write $-\omega$ and $-\omega^2$ in exponential form.

Model Answer

$$-\omega = e^{-i\pi/3} = \frac{1 - i\sqrt{3}}{2}, \qquad -\omega^2 = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}$$

These are the sixth roots of unity; $(-\omega)^6 = 1$. Ex: $(1 + i\sqrt{3}) = -2\omega^2$, so $(1 + i\sqrt{3})^p + (1 - i\sqrt{3})^p = (-2)^p(\omega^{2p} + \omega^p)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 40; J-A 29; C-II 3 | DPP: 4(10); 6(14)

Question 82
What are the roots of the equations $x^2 + x + 1 = 0$ and $x^2 - x + 1 = 0$?

Model Answer

$$x^2 + x + 1 = 0 : \ x = \omega, \omega^2; \qquad x^2 - x + 1 = 0 : \ x = -\omega, -\omega^2$$

Ex: roots of $x^2 - x + 1 = 0$: $\alpha^{101} + \beta^{107} = (-\omega)^{101} + (-\omega^2)^{107} = -\omega^2 - \omega = 1$.

Grade - Low
Practice map, main concept: Sheet: J-I 43
Also used in: Sheet: C-II 17 | DPP: 5(6, 10)

n^{th} Roots of Unity

Question 83
What are the nth roots of unity and how are they represented?

Model Answer

$$z^n = 1 : \ z = e^{i2k\pi/n}, \ k = 0, 1, \ldots, n-1; \ \text{i.e. } 1, \alpha, \alpha^2, \ldots, \alpha^{n-1}, \ \alpha = e^{i2\pi/n}$$

Ex: fourth roots: $1, i, -1, -i$.

Grade - Medium
Practice map, main concept: Sheet: J-A 2 | DPP: 1(8)
Also used in: none

Question 84
What is the sum of all n^{th} roots of unity?

Model Answer

$$1 + \alpha + \alpha^2 + \cdots + \alpha^{n-1} = 0 \quad (n \geq 2)$$

Ex: $\sum_{k=1}^{6} \left(\cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7} \right) = -1$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 44; C-II 16

Question 85
What is the product of all n^{th} roots of unity?

Model Answer

$$1 \cdot \alpha \cdot \alpha^2 \cdots \alpha^{n-1} = (-1)^{n-1}$$

Ex: $n = 4$: $1 \cdot i \cdot (-1) \cdot (-i) = -1 = (-1)^3$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 5(13)

Question 86
What is the common ratio of the GP formed by all n^{th} roots of unity?

Model Answer
Common ratio $\alpha = e^{i2\pi/n}$. Common roots of $1 + z + \cdots + z^{17} = 0$ and $1 + z + \cdots + z^{13} = 0$: $z^{18} = 1 = z^{14}$, $z \neq 1$, so $z^2 = 1$, $z = -1$ only.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 1(8)

Question 87
How to find the sum of any r^{th} power of all n^{th} roots of unity?

Model Answer

$$\sum_{k=0}^{n-1} \alpha^{kr} = \begin{cases} n, & n \mid r \\ 0, & n \nmid r \end{cases}$$

Ex: eighth roots: $\sum \alpha_k^3 = 0$, $\sum \alpha_k^8 = 8$.

Grade - Low
Practice map, main concept: Sheet: J-I 44; C-II 16 | DPP: 5(11-13); 6(11)
Also used in: DPP: 6(5)

Question 88
What do all n^{th} roots of unity represent on the complex plane, and can you find the dimensions of that polygon?

Model Answer
Vertices of a regular n-gon inscribed in $|z| = 1$: side $2\sin\dfrac{\pi}{n}$, area $\dfrac{n}{2}\sin\dfrac{2\pi}{n}$.

Ex: roots of $z^6 = 1$ form a regular hexagon of side 1 and area $\dfrac{3\sqrt{3}}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-A 9; C-II 19
Also used in: DPP: 5(3)

Question 89
What are the factors of $(x^n - 1)$ involving n^{th} roots of unity?

Model Answer

$$x^n - 1 = (x-1)(x-\alpha)(x-\alpha^2)\cdots(x-\alpha^{n-1}), \qquad \prod_{k=1}^{n-1}(x-\alpha^k) = 1 + x + \cdots + x^{n-1}$$

Ex: $\prod_{k=1}^{n-1}(2-\alpha^k) = 2^n - 1$; $\prod_{k=1}^{n-1}|1-\alpha^k| = n$.

Grade - Medium

Practice map, main concept: Sheet: J-A 7, 26; C-I 23, 26 | DPP: 6(5)
Also used in: Sheet: J-II 35

Question 90
What are the factors of $(x^n - 1)$ without involving n^{th} roots of unity?

Model Answer

Pair each root with its conjugate: $(x-\alpha^k)(x-\alpha^{n-k}) = x^2 - 2x\cos\dfrac{2k\pi}{n} + 1$ (real quadratic factors).

Ex: $z^5 - 32 = (z-2)\left(z^2 - 4z\cos\dfrac{2\pi}{5} + 4\right)\left(z^2 - 4z\cos\dfrac{4\pi}{5} + 4\right)$.

Grade - Low

Practice map, main concept: Sheet: J-II 35; C-II 9
Also used in: none

Question 91
What is the modulus of any n^{th} root of unity?

Model Answer

$|\alpha^k| = 1$; all lie on the unit circle, so $\bar{\alpha}^k = \dfrac{1}{\alpha^k}$. Distance between consecutive roots $= 2\sin\dfrac{\pi}{n}$.

Ex: $|\alpha_{k+1} - \alpha_k| = 2\sin\dfrac{\pi}{14}$ for the 14th roots $\alpha_k = e^{ik\pi/7}$.

Grade - Medium

Practice map, main concept: none
Also used in: Sheet: J-A 9, 26

Question 92
Which two n^{th} roots of unity are conjugates of each other?

Model Answer

$$\overline{\alpha^k} = \alpha^{n-k}$$

Ex: $n = 5$: $\bar{\alpha} = \alpha^4$, $\overline{\alpha^2} = \alpha^3$.

Grade - Medium

Practice map, main concept: none
Also used in: Sheet: J-A 7 | DPP: 5(11)

Question 93
The product of which two n^{th} roots of unity is 1?

Model Answer

$$\alpha^k \cdot \alpha^{n-k} = \alpha^n = 1$$

Ex: $\alpha^5 = 1$, $\alpha \neq 1$: $1 + \alpha + \alpha^2 + \alpha^{-2} - \alpha^{-1} = 1 + \alpha + \alpha^2 + \alpha^3 - \alpha^4 = -2\alpha^4$, modulus 2.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 7 | DPP: 6(11)

Summation of Series using Complex Numbers

Question 94
Prove the result of summation of a series of the type $\sum$(trig ratio) using complex numbers.

Model Answer
Form $C + iS = \sum e^{i(\alpha+k\beta)}$ (a GP), sum it, and take the real / imaginary part:

$$\sum_{k=0}^{n-1} \cos(\alpha + k\beta) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \cos\left(\alpha + \frac{(n-1)\beta}{2}\right)$$

Ex: $\sum_{m=1}^{15} \sin(2m-1)2^\circ = \dfrac{\sin^2 30^\circ}{\sin 2^\circ} = \dfrac{1}{4\sin 2^\circ}$.

Grade - Low

Practice map, main concept: Sheet: J-I 13; J-II 34

Also used in: Sheet: C-II 13

Complex Numbers and Binomial Coefficients

Question 95
How to solve questions of series based on binomial coefficients, and when to put i, when to put ω?

Model Answer
In $(1+x)^n = \sum C_r x^r$ put $x = i$ when the terms change sign in pairs / skip by 2 with alternate signs, $x = \omega$ when suffixes skip by 3; then compare real and imaginary parts.

$$C_0 - C_2 + C_4 - \cdots = 2^{n/2}\cos\frac{n\pi}{4}, \qquad C_1 - C_3 + C_5 - \cdots = 2^{n/2}\sin\frac{n\pi}{4}$$

Grade - Medium

Practice map, main concept: Sheet: C-II 13 | Question Bank: 347

Also used in: none

Question 96
How to find the sum of coefficients of all powers of the form x^{3m}, x^{3m+1} and x^{3m+2} in any expansion?

Model Answer

$$\sum a_{3m} = \frac{f(1) + f(\omega) + f(\omega^2)}{3}, \quad \sum a_{3m+1} = \frac{f(1) + \omega^2 f(\omega) + \omega f(\omega^2)}{3}, \quad \sum a_{3m+2} = \frac{f(1) + \omega f(\omega) + \omega^2 f(\omega^2}{3}$$

Ex: $C_0 + C_3 + C_6 + \cdots = \dfrac{1}{3}\left(2^n + 2\cos\dfrac{n\pi}{3}\right)$; for $n = 6$: $1 + 20 + 1 = 22$.

Grade - Low

Practice map, main concept: Sheet: C-II 11, 26
Also used in: none

Question 97
How to find the sum of coefficients of all powers of the form x^{pm}, x^{pm+1}, ..., $x^{pm+(p-1)}$ in any expansion?

Model Answer

$$\sum_m a_{pm+r} = \frac{1}{p}\sum_{k=0}^{p-1} \alpha^{-kr} f(\alpha^k), \qquad \alpha = e^{i2\pi/p}$$

Ex ($p=4$): $C_1 + C_5 + C_9 + \cdots = \frac{1}{4}\left(2^n + 2^{n/2+1}\sin\frac{n\pi}{4}\right)$; for $n=5$: $5+1=6$.

Grade - Low
Practice map, main concept: Sheet: C-II 27-28
Also used in: none

Straight Line in Complex Numbers

Question 98
How to assume any complex number on the line joining two given points?

Model Answer
$z = z_1 + t(z_2 - z_1) = (1-t)z_1 + tz_2$, $t \in \mathbb{R}$ ($0 \le t \le 1$ for the segment).
Equivalently $\dfrac{z - z_1}{z_2 - z_1}$ is real. Ex: on the segment from $-3+5i$ to $-5-3i$ the argument of z varies from $\pi - \tan^{-1}\dfrac{5}{3}$ to $-\pi + \tan^{-1}\dfrac{3}{5}$ through π.

Grade - Medium
Practice map, main concept: Sheet: J-II 43, 47 | DPP: 3(13)
Also used in: Sheet: J-M 19 | DPP: 5(8) | Question Bank: 342

Question 99
What is the general form of the equation of a line in the complex plane, and how to convert it into the form $ax + by + c = 0$?

Model Answer
$\bar{a}z + a\bar{z} + b = 0$ ($b \in \mathbb{R}$, $a \neq 0$); with $a = p + iq$: $2(px + qy) + b = 0$, slope $-\dfrac{p}{q}$.

Through z_1, z_2: $\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$. Ex: $(1+2i)z + (2i-1)\bar{z} = 10i$ with $z = x+iy$ gives $i(4x+2y) = 10i$, i.e. $2x + y = 5$ (imaginary-axis intercept 5).

Grade - High
Practice map, main concept: Sheet: J-II 25, 50; J-M 18, 32, 39, 41, 43; J-A 14; C-I 7, 27 | DPP: 6(3) | Question Bank: 317, 324, 326, 335, 341, 345, 356
Also used in: Sheet: J-A 4; C-II 23

Question 100

How to find the image of a point when the equation of the line is given in general form?

Model Answer

Image z_2 of z_1 in $\bar{a}z + a\bar{z} + b = 0$: $\bar{a}z_2 + a\bar{z}_1 + b = 0$, so $z_2 = -\dfrac{a\bar{z}_1 + b}{\bar{a}}$.

Ex: image of $z_1 = 2 + i$ in $z + \bar{z} = 0$ (imaginary axis, $a = 1, b = 0$): $z_2 = -\bar{z}_1 = -2 + i$.

Grade - Low

Practice map, main concept: none

Also used in: none

Circle in Complex Numbers

Question 101

What is the general form of the equation of a circle in the complex plane, and how to convert it into the general form studied in the chapter Circle?

Model Answer

$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$ ($b \in \mathbb{R}$): centre $-a$, radius $\sqrt{|a|^2 - b}$.

Put $z = x+iy$, $a = p+iq$: $x^2+y^2+2px+2qy+b = 0$. Diametric form: $|z-z_1|^2+|z-z_2|^2 = |z_1-z_2|^2$.

Ex: $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + 5 = 0$: $a = 4 + 3i$, radius $\sqrt{25 - 5} = 2\sqrt{5}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 19; J-II 49; J-M 23, 25; C-II 18 | DPP: 3(12); 6(4)

Also used in: Sheet: J-A 19; C-I 7, 27 | Question Bank: 313

Question 102

How to check whether 4 given points on the complex plane are concyclic?

Model Answer

z_1, z_2, z_3, z_4 (in order) are concyclic $\iff \dfrac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)}$ is purely real.

Ex: $1, i, -1, -i$: $\dfrac{(-i - 1)(i + 1)}{(i - 1)(-i + 1)} = \dfrac{-2i}{2i} = -1$ (real), so concyclic.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 13